

MATH 3191 Written Homework 3

____/20 points

Name: Key

Due: September 16th

Show all work that leads to your answers. I will be providing feedback on the work as well as the answers!

1. (10 points) Determine if the following 3 vectors are linearly independent.

$$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Solution: There are many ways to solve this one. Since we are going to invert a matrix whose columns are the above vectors, we will go ahead and try to compute the inverse here to save computation

$$\begin{aligned} [A \mid I] &= \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 3 & -1 & -1 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -10 & -4 & -3 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 + 5R_2} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -3 & 5 & 1 \end{array} \right] \\ &\xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -3 & 5 & 1 \end{array} \right] \\ &\xrightarrow{R_2 \leftarrow R_2 - \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{3}{2} & -2 & -\frac{1}{2} \\ 0 & 0 & 1 & -3 & 5 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 - R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 4 & -5 & -1 \\ 0 & 1 & 0 & \frac{3}{2} & -2 & -\frac{1}{2} \\ 0 & 0 & 1 & -3 & 5 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{3}{2} & -2 & -\frac{1}{2} \\ 0 & 0 & 1 & -3 & 5 & 1 \end{array} \right] \end{aligned} \tag{1}$$

Since we were able to compute the inverse, the columns of the original matrix we computed were linearly independent as desired.

Note that we could have stopped at 1 if we were only interested in if these three vectors were linearly independent. Since we have 3 pivot columns, we know these three are linearly independent. However, as we stated above, we will want the inverse for a later problem, so we keep going!

2. (10 points) Let $A \in \mathbb{R}^{3 \times 3}$ such that

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 1 \\ 3 & -1 & -1 \end{bmatrix}$$

Solve the following system for $\mathbf{x}$ by using A^{-1}

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 1 \\ 3 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \\ -11 \end{bmatrix}$$

Solution: Since we computed the inverse of the above matrix in problem 1, we just have to multiply both sides by A^{-1} .

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -\frac{1}{2} & 1 & \frac{1}{2} \\ \frac{3}{2} & -2 & -\frac{1}{2} \\ -3 & 5 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \\ -11 \end{bmatrix} \\ &= 7 \begin{bmatrix} -\frac{1}{2} \\ \frac{3}{2} \\ -3 \end{bmatrix} + 9 \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} - 11 \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{7}{2} + 9 - \frac{11}{2} \\ \frac{21}{2} - 18 + \frac{11}{2} \\ -21 + 45 - 11 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ -2 \\ 13 \end{bmatrix} \end{aligned}$$